FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014: 2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

ALL INDIA INTEGRATED TEST SERIES

FIITJEE

JEE(Advanced), 2015

ANSWERS, HINTS & SOLUTIONS

HALF COURSE TEST –VIII
(Paper-2)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICCS
1.	**B**	**B**	**B**
2.	**C**	**C**	**C**
3.	**A**	**D**	**C**
4.	**A**	**A**	**D**
5.	**A, C**	**A, B, C**	**D**
6.	**B, C**	**A, C, D**	**A, B, C**
7.	**A, B, C**	**A, B, C**	**A, C, D**
8.	**A, C**	**A, B, C**	**B, C**
9.	**B, C**	**A, B, C**	**A, D**
1.	**(A) → (t)** **(B) → (q)** **(C) → (q)** **(D) → (p)**	**(A) → (q, r, t)** **(B) → (p, r)** **(C) → (p, r, s, t)** **(D) → (p, r)**	**(A) → (p, q, r, s)** **(B) → (q)** **(C) → (r)** **(D) → (p)**
2	**(A) → (p, q, r, s, t)** **(B) → (p, q, r, s, t)** **(C) → (p, q, r, s, t)** **(D) → (r)**	**(A) → (p, q)** **(B) → (p, q, s)** **(C) → (p, q, s)** **(D) → (q, r, s, t)**	**(A) → (r)** **(B) → (s)** **(C) → (q)** **(D) → (p)**
1.	**3**	**3**	**8**
2.	**8**	**4**	**3**
3.	**6**	**3**	**8**
4.	**8**	**4**	**6**
5.	**5**	**6**	**1**
6.	**2**	**4**	**3**
7.	**8**	**2**	**9**
8.	**2**	**4**	**5**

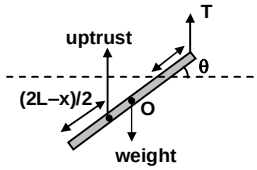
Physics

PART – I

SECTION – A

1. $\omega_0 = \sqrt{\frac{K}{m}}$, $\omega' = \sqrt{\frac{8K}{m/2}} = 4\,\omega_0$

2. $$E_1 = -\frac{GM_e m}{R_e + \frac{R_e}{2}} - \left(-\frac{GM_e m}{R_e}\right) = \frac{GM_e m}{3R_e}$$
$$E_2 = \frac{1}{2}mv_0^2 = \frac{1}{2}m.\frac{GM_e}{R + \frac{R_e}{2}} = \frac{GM_e m}{3R_e}.$$
$\therefore$ (C)

3. For vertical equilibrium
T = weight – upthrust
$$T = 2LA\left(\frac{3}{4}\right)\rho g - (2L - x)A\rho g = Ag\rho\left[x - \frac{L}{2}\right]$$


where ρ is the density of water
for rotational equilibrium about O
$$(2L - x)Ag\rho\left[L - \left(\frac{2L - x}{2}\right)\right]\cos\theta = Ag\rho\left[x - \frac{L}{2}\right]L\cos\theta$$
$\Rightarrow$ x = L

4. If v be the velocity at mean position in two cases then
$$\frac{1}{2}mv^2 = \frac{1}{2}kA_1^2 \text{ and } T_1 = T_2 = 2\pi\sqrt{\frac{m}{k}}$$
$A_1 = A_2$

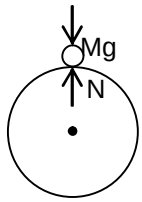
5. Buoyant forces on both the spheres will be same because volume of both the spheres is same and weight of the hollow sphere is less. If 'A' and 'a' are the accelerations of solid and hollow spheres respectively, then A > a.

6. $P = \sigma(4\pi r^2)T^4$
$$R = \frac{dT}{dt} = \left(\frac{3\sigma T^4}{\rho s}\right)\frac{1}{r}$$

SECTION – C

1. At point P slope is tan 150°
$$\therefore \quad \frac{d\rho}{dT} = \tan 150° = -\frac{1}{\sqrt{3}}$$
Since $\rho = \frac{PM}{RT} \Rightarrow \frac{d\rho}{dT} = -\frac{PM}{RT^2}$

3. $\frac{V^2}{R} = 4ms^{-2}$



$Mg - N = 10 - N = 1.\frac{V^2}{R} = 4$

$\Rightarrow$ N = 6 Newtons

4. $\omega = \sqrt{\frac{K}{M}} \quad T = 2\pi\sqrt{\frac{M}{K}}$

$T_1 = 2\pi\sqrt{\frac{M_1M_2}{K(M_1+M_2)}}$ $\qquad$ $T_2 = 2\pi\sqrt{\frac{M_2}{K}}$

$T_2 = 2\pi\sqrt{\frac{M_2}{K}}$

$\frac{M_1+M_2}{M_1} = \left(\frac{T_2}{T_1}\right)^2 = 9 = \frac{M_1+M_2}{M_1} = \frac{1+M_2}{M_1}$

$\therefore \quad \frac{M_2}{M_1} = 8$

5 $\omega = 120\,\pi$

K = $2\pi / 3$

$V = \frac{\omega}{K} = 180 = 36X$

X = 5

6. While floating net force = zero

Increase in buoyancy force = increase in weight

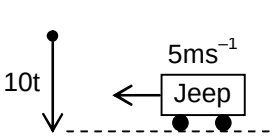
$\frac{V}{2}\rho g = x\rho g$

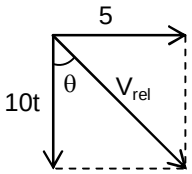
$\Rightarrow x = \frac{V}{2}$

7. $\tan\theta = \frac{\ell}{Mg} = \left[\frac{15.92 - 7.96}{1\times 10}\right]10^{-6} = 7.96\times 10^{-7}\,\frac{m}{Newton}$

$Y = \frac{2m}{3.14(10^{-3}m)^2}\,\frac{N}{7.96\times 10^{-7}m} = 8\times 10^{-11}\,\frac{N}{m^2}$

8.





$V_{rel} = 5\sqrt{1+4t^2}$

$\sin\theta = \frac{5}{5\sqrt{1+4t^2}} = \frac{1}{\sqrt{1+4t^2}}$ $\qquad$ $g\sin\theta = a_N = \frac{10}{\sqrt{1+4t^2}}$

$R = \frac{V_{rel^2}}{a_N} = \frac{25}{10}(1+4t^2)^{3/2} = \frac{5}{2}(17)^{3/2}$ $\qquad$ $\Rightarrow 1+4t^2 = 17$

t = 2 seconds.

Chemistry

PART – II

SECTION – A

3. (i) use PV = nRT

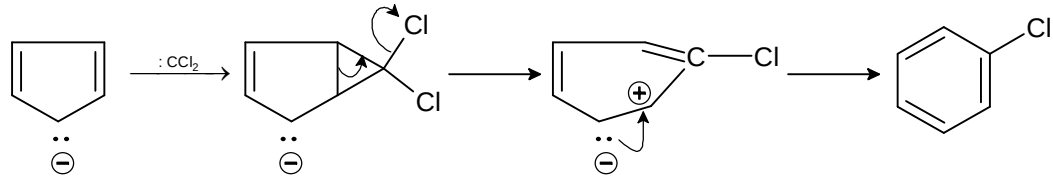
$PV = \frac{w}{m}RT$

(ii) due to hydrogen bonding $a_{NH_3} > a_{CO_2}$

(iii) mean free path $(\lambda) \propto \frac{1}{\sigma^2}$

(iv) Average K.E per molecule = $\frac{3}{2}$ kT.

5. (A) Aniline does not undergo Fridel Craft reaction due to lone pair on N-atom which directly attacks on $AlCl_3$.
(B) formation of carbocation at bridge head position is not favoured.
(C) $CHCl_3 + \text{alc. KOH} \longrightarrow :CCl_2$ (dichlorocarbene)



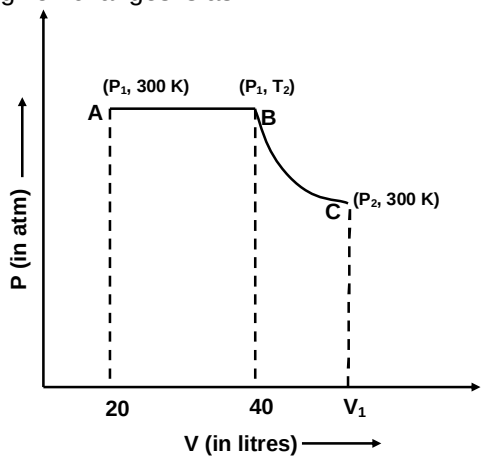
(D) Three times protonation followed by ring expansion gives benzene.

6. In the option (b), the H atom is linked to an sp^2 hybridised C atom.

8. $r = K [A]^m [B]^n$
on calculation, m = 2, n = 1 & $K = 2.66 \times 10^8$ $mol^{-2} l^2 sec^{-1}$ at 300 K
so, K will be > 2.66×10^8 at 320 K.
Hence, (A, B, C)

SECTION – C

1. The PV diagram for the given changes is as:



for isobaric process AB,

$\frac{V_1}{T_1} = \frac{V_2}{T_2}$ (from Charle's law)

$\frac{20}{300} = \frac{40}{T_2}$,

$\therefore \quad T_2 = 600$ K

$w_t = w_{AB} + w_{BC}$

Now, w_{AB} = nRdT (Isobaric process)

$= -2 \times 2 \times (600 - 300) = -1200$ cal.

w_{BC} = for adiabatic process

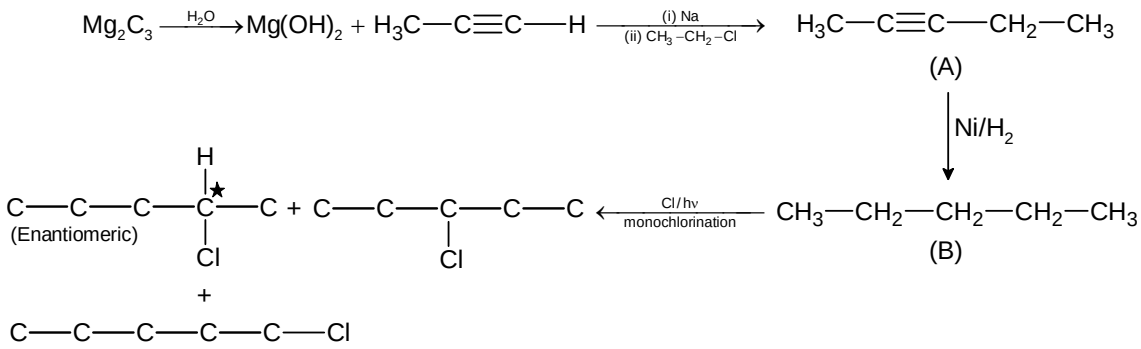
$w = -n\, C_V \Delta T$

$= -2 \times \frac{3}{2} \times R \times (600 - 300)$

$= -1800$ Cal

$\therefore w_T = -1200 - 1800 = -3000$ Cal $= -3$ kcal.

2.



3. 30 ml of N/50 HCl = 30 ml of N/50 $Mg(HCO_3)_2$

= 30 ml of N/50 $CaCO_3$ = 100 ml of tap water

$\therefore$ mass of $CaCO_3$ in 100 ml tap water = $\frac{E \times N \times V}{1000}$

$= \frac{50 \times 30}{50 \times 1000} = 0.03$ g

$\therefore$ Hardness = 300 ppm

X = 3

4. $NaBH_4 + I_2 \xrightarrow[\text{solution}]{\text{diglyme}} \underset{(X)}{B_2H_6} + H_2 + 2NaI$

$B_2H_6 + 2NH_3 \longrightarrow \underset{(\text{adduct})}{[B_2H_6.2NH_3]} \xrightarrow{\Delta} \underset{(Y)}{B_3N_3H_6} \xrightarrow{Cl_2} \underset{(\text{dichloro derivatives})}{B_3N_3H_4Cl_2}$

So, dichloro derivatives of (Y) = 4

Mathematics

PART – III

SECTION – A

1. Let the equation of the line L be $y - 2 = m(x - 8)$, $m < 0$

$\therefore P = \left(8 - \frac{2}{m}, 0\right)$, $Q = (0, 2 - 8m)$

$\therefore OP + OQ = 8 - \frac{2}{m} + 2 - 8m \geq 18$

2. Let centre C = (a, 0)

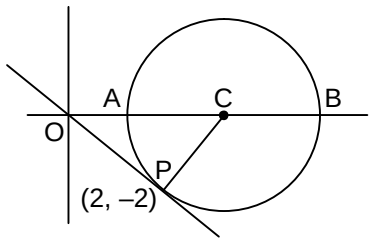
$\angle COP = 45^\circ$

$OP = 2\sqrt{2} = CP$

$\therefore OC = 4$

The point on the circle with greatest abscissa is B

$\alpha = OB = 4 + 2\sqrt{2}$



3. $y = mx - 2am - am^3$, $y = mx + 2b + \frac{b}{m^2}$

For common normal, $2b + \frac{b}{m^2} + 2am + am^3 = 0$

$\Rightarrow am^5 + 2am^3 + 2bm^2 + b = 0$

$\therefore$ Maximum 5 normal

4. Equation of tangent $y = mx + \sqrt{a^2m^2 + b^2}$

Equation of a perpendicular from centre (0, 0) on the tangent is $y = -\frac{1}{m}x$

$\therefore$ Required locus $y = -\frac{x}{y}x + \sqrt{a^2\frac{x^2}{y^2} + b^2}$

5. Pair of tangents $SS' = T^2$

$\tan\theta = \frac{2\sqrt{h^2 - ab}}{a + b} = \frac{2\sqrt{(a^2 - 1)}}{a^2 - 2}$

$\tan\theta < 0$

$\therefore \frac{2\sqrt{a^2 - 1}}{a^2 - 2} < 0$

$\Rightarrow a \in (-\sqrt{2}, -1) \cup (1, \sqrt{2})$

6. $2s(2s - 2a) = xbc$

$\Rightarrow x = \frac{4s(s - a)}{bc} = 4\cos^2\frac{A}{2}$

$\therefore x \in (0, 4)$

7. $x^2 + x + 1 = ax + 1$ and $-1 \leq x^2 + x + 1 \leq 1$

$\Rightarrow \theta \leq a < 1$

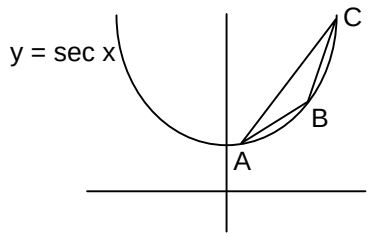
8. $\cos^4\theta - \sin^4\theta = \cos^2\theta - \sin^2\theta = \cos 2\theta$
$\Rightarrow (-2b)^2 - 4b = (-4)^2 - 4 \times 2$
$\therefore$ b = –1, 2

9. Put x = 1
$\Rightarrow (2 + 3c + c^2)^{12} = 0$ (as the sum of coefficients vanishes)
$\Rightarrow c^2 + 3c + 2 = 0$
$\Rightarrow$ c = –1, –2

SECTION – B

1. (A) Can take any value except $\left(\frac{1}{3}, 3\right)$

(C) Centroid of ΔABC is $\left(\frac{A+B+C}{3}, \frac{\sec A + \sec B + \sec C}{3}\right)$



$\therefore \frac{\sec A + \sec B + \sec C}{3} \geq \sec\left(\frac{A+B+C}{3}\right)$

$\frac{\sec A + \sec B + \sec C}{3} \geq 2 \Rightarrow$ Least value of $\sum \sec A = 6$

(D) $\frac{2\sin 20^\circ \cos 20^\circ}{\frac{1}{2} \times \sin 20^\circ \cos 20^\circ} = 4$

2. R = 15

$r = \frac{\Delta}{s} = \frac{\frac{1}{2} \times 18 \times 24}{\frac{18+24+30}{2}} = 6$

Circum-centre = (9, 12), orthocentre = (0, 0), in-centre = (6, 6), centroid = (6, 8)

SECTION – C

1. $x^2 + y^2 - 12x - 2y + 28 = 0$
C = (6, 1), r = 3
$\therefore |z_0 + 5 - i|_{min} = 8$

2. $|z - 1| + |z - 2| + |z - 3| \leq |3z - 6| = 3|z - 2| = 3|2e^{i\theta} - 2| = 12$

3. $k = 2^9$
$\Rightarrow na + \frac{n(n-1)}{2} = 2^9$
$\Rightarrow n(2a + n - 1) = 2^{10}$
Which is only possible for n = 1
$\Rightarrow$ a = 512

4. 6 boys and 5 girls in $\frac{11!}{6!\,5!}$

Teachers are at remaining places in ${}^{12}P_3$ ways

$\therefore\ {}^{12}P_3 \times \frac{11!}{5!\,6!}$

5. $b \geq \sqrt{ac}$

$\Rightarrow b^3 \geq abc$

$\Rightarrow b^3 \geq 4$

$\Rightarrow b \geq 4^{1/3}$

$\therefore [b] = 1$

6. $6 \cot\theta = \cot(\theta - \alpha) + \cot(\theta + \alpha) = \frac{\sin 2\theta}{\sin(\theta+\alpha)\sin(\theta-\alpha)}$

$\Rightarrow \frac{3\cos\theta}{\sin\theta} = \frac{2\sin\theta\cos\theta}{\cot 2\alpha - \cos 2\theta}$

$\Rightarrow 4\sin^2\theta = 6\sin^2\alpha$

$\therefore \frac{2\sin^2\theta}{\sin^2\alpha} = 3$

7. $x = y = z = 1$

$\therefore 2x + 3y + 4z = 9$

8. $2z\bar{z} + 2 \mid z-3 \mid\mid z+3 \mid = 82$

$\therefore |z + 3| + |z - 3| = 10$

$\therefore |z|_{max} = 5$